Chapter 20 Carbohydrate Biosynthesis in Plants and Bacteria

Multiple Choice Questions

1. Which of the following is *not* true about plastids?

   A) Most of the proteins in a plastid are encoded in nuclear genes.
   B) Plastids are encoded by a double membrane.
   C) Plastids are self-replicating.
   D) Plastids contain a small circular genome.
   E) The inner membrane of plastids is permeable to polar and charged molecules.

Photosynthetic carbohydrate synthesis
Pages: 800–801 Difficulty: 1 Ans: E

2. The compound that condenses with CO_2 in the first reaction of carbon dioxide assimilation is:

   A) 3-phosphoglycerate.
   B) ribose 1,5-bisphosphate.
   C) ribulose 1,5-bisphosphate.
   D) ribulose 5-phosphate.
   E) rubisco.

Photosynthetic carbohydrate synthesis
Page: 802 Difficulty: 1 Ans: C

3. The final product that is formed by the enzyme rubisco is:

   A) 3-phosphoglycerate.
   B) ATP.
   C) ribulose 1,5-bisphosphate.
   D) glyceraldehyde-3-phosphate.
   E) CO_2.

Photosynthetic carbohydrate synthesis
Pages: 802-803 Difficulty: 1 Ans: A

4. Which of the following is *not* required to activate rubisco?

   A) ATP hydrolysis
   B) Carbamoylation of the active site lysine by rubisco activase
   C) Release of ribulose 1,5-bisphosphate from the active site
   D) Release of 2-carboxyarabinitol from the active site
   E) Binding of Mg^{2+}

Photosynthetic carbohydrate synthesis
Page: 804 Difficulty: 2 Ans: B

5. Which of these enzymes is *not* part of the Calvin cycle?

A) Aldolase
B) Glyceraldehyde 3-phosphate dehydrogenase
C) Phosphofructokinase-1
D) Ribulose-5-phosphate kinase
E) Transketolase

Photosynthetic carbohydrate synthesis
Pages: 801–806 Difficulty: 2 Ans: C

6. Transketolase requires the coenzyme:

A) cobalamin (vitamin B_{12}).
B) lipoic acid
C) pyridoxal phosphate.
D) tetrahydrofolic acid.
E) thiamine pyrophosphate.

Photosynthetic carbohydrate synthesis
Pages: 805–807 Difficulty: 2 Ans: E

7. When transketolase acts on fructose 6-phosphate and glyceraldehyde 3-phosphate, the products are:

A) 3-phosphoglycerate and glyceraldehyde 3-phosphate.
B) 3-phosphoglycerate and two molecules of glyceraldehyde 3-phosphate.
C) dihydroxyacetone phosphate and glucose 6-phosphate.
D) xylulose 5-phosphate and erythrose 4-phosphate.
E) xylulose 5-phosphate and ribose 5-phosphate.

Photosynthetic carbohydrate synthesis
Pages: 805–807 Difficulty: 2 Ans: D

8. Which of these compounds is *not* directly involved in the Calvin cycle?

A) Erythrose 4-phosphate
B) Glyceraldehyde 3-phosphate
C) Mannose 6-phosphate
D) Ribulose 5-phosphate
E) Sedoheptulose 7-phosphate

Photosynthetic carbohydrate synthesis
Pages: 801–806 Difficulty: 1 Ans: C

9. In the carbon assimilation ("dark") reactions of photosynthesis, the biosynthesis of 1 mol of hexose from 6 mol of carbon dioxide requires:

A) 12 mol of NADPH and 12 mol of ATP.
B) 12 mol of NADPH and 18 mol of ATP.
C) 18 mol of NADPH and 12 mol of ATP.
D) 18 mol of NADPH and 18 mol of ATP.

E) no NADPH and 12 mol of ATP.

Photosynthetic carbohydrate synthesis
Pages: 808–809 Difficulty: 2 Ans: B

10. The known mechanisms of activation of rubisco or of other enzymes of the Calvin cycle during illumination include all of the following *except*:

A) increased stromal pH.
B) light-driven entry of Mg^{2+} into the stroma.
C) phosphorylation by cAMP-dependent protein kinase.
D) phosphorylation of phosphoenolpyruvate carboxylase.
E) reduction of a disulfide bridge by thioredoxin.

Photosynthetic carbohydrate synthesis
Pages: 810–811 Difficulty: 3 Ans: C

11. Which of these chloroplast enzymes is *not* regulated by light?

A) Fructose 1,6-bisphosphatase
B) Glyceraldehyde-phosphate dehydrogenase
C) Ribulose 5-phosphate kinase
D) Sedoheptulose 1,7-bisphosphatase
E) All of the above

Photosynthetic carbohydrate synthesis
Pages: 810–811 Difficulty: 3 Ans: E

12. The carbon assimilation ("dark") reactions of photosynthetic plants:

A) are driven ultimately by the energy of sunlight.
B) are important to plants, but ultimately of little significance for bacteria and animals.
C) cannot occur in the light.
D) yield (reduced) NADH.
E) yield ATP, which is required for the light reactions.

Photosynthetic carbohydrate synthesis
Pages: 808–809 Difficulty: 2 Ans: A

13. The assimilation of CO_2 into organic compounds (triose phosphates) in green plants:

A) involves condensation of the two-carbon compound acetate with CO_2 to form 3-phosphoglycerate.
B) requires NADPH.
C) results in the production of ATP.
D) takes place at equal rates in light and darkness.
E) takes place in the cytosol.

Photosynthetic carbohydrate synthesis
Pages: 805–808 Difficulty: 2 Ans: B

14. Which of the following is *not* true of the P_i-triose phosphate antiporter?

A) The antiporter can move phosphate in or out of the chloroplast.
B) When P_i is moved into the chloroplast, it supports synthesis of ATP.
C) When triose phosphates are moved out of the chloroplast, they become incorporated into sucrose.
D) ATP and reducing equivalents are moved into the chloroplast by the antiporter.
E) The antiporter is found in the inner membrane of chloroplasts.

Photosynthetic carbohydrate synthesis
Pages: 809–810 Difficulty: 1 Ans: D

15. All are true of photorespiration *except*:

A) It is driven by light.
B) It oxidizes substrates to CO_2.
C) It produces O_2.
D) It results from a lack of specificity of the enzyme rubisco.
E) It results in no fixation of carbon.

Photorespiration and the C_4 and CAM pathways
Pages: 812–813 Difficulty: 1 Ans: C

16. The three subcellular organelles involved in the phosphoglycolate salvage pathway are:

A) endoplasmic reticulum, chloroplast, and mitochondrion.
B) nucleus, endoplasmic reticulum, and chloroplast.
C) golgi apparatus, chloroplast, and mitochondrion.
D) mitochondrion, peroxisome, and chloroplast.
E) peroxisome, endoplasmic reticulum, and chloroplast.

Photorespiration and the C_4 and CAM pathways
Page: 813 Difficulty: 1 Ans: D

17. The glycine decarboxylase complex in the leaves of pea or spinach plants is localized mainly in the:

A) chloroplast.
B) endoplasmic reticulum.
C) mitochondrion.
D) cell membrane.
E) peroxisome.

Photorespiration and the C_4 and CAM pathways
Page: 813 Difficulty: 1 Ans: C

18. In "C_4" plants of tropical origin, the first intermediate into which $^{14}CO_2$ is fixed is:

A) aspartate.
B) phosphoenolpyruvate.
C) oxaloacetate.
D) malate.
E) 3-phosphoglycerate.

Photorespiration and the C_4 and CAM pathways
Page: 815 Difficulty: 1 Ans: C

19. The synthesis of glycogen, starch, and sucrose all:

A) involve addition of a sugar residue at the reducing end of the growing polymer.
B) take place in liver and muscle of mammals.
C) use a sugar nucleotide as substrate.
D) use glucose 1-phosphate as the only substrate.
E) use glucose-6-phosphate as substrate.

Biosynthesis of starch and sucrose
Page: 819 Difficulty: 2 Ans: C

20. The synthesis of starch and sucrose in plants uses _____ as the substrate, rather than _____, which is used in the synthesis of glycogen in animal cells.

A) ADP-fructose; UDP-glucose
B) ADP-glucose; UDP-glucose
C) fructose 1-phosphate; glucose 1-phosphate
D) glucose 1-phosphate; glucose 6-phosphate
E) UDP-glucose; ADP-glucose

Biosynthesis of starch and sucrose
Page: 819 Difficulty: 2 Ans: B

21. The precursors for sucrose biosynthesis are:

A) glucose and fructose.
B) UDP-glucose and fructose 6-phosphate.
C) UDP-fructose and glucose 6-phosphate.
D) UDP-glucose and fructose.
E) UDP-glucose and UDP-fructose.

Biosynthesis of starch and sucrose
Page: 819 Difficulty: 2 Ans: B

22. The major regulator of sucrose biosynthesis in plants is:

A) fructose 2,6-bisphosphate.
B) fructose 1,6-bisphosphate.
C) sucrose.
D) glucose and fructose.
E) glucose 6-phosphate.

Biosynthesis of starch and sucrose
Page: 820 Difficulty: 2 Ans: A

23. Which of the following is *not* true about the biosynthesis of cellulose?

A) UDP-glucose is used as a substrate.

B) Glucose is transported across the membrane via a lipid-linked oligosaccharide intermediate.
C) The extracellular form of cellulose synthase adds up to 15,000 glucose units to one chain.
D) Cellulose polymers line up in an anti-parallel fashion.
E) The reaction catalyzed by cellulose synthase proceeds via inversion of configuration.

Synthesis of cell wall polysaccharides: plant cellulose and bacterial peptidoglycan
Pages: 821–823 Difficulty: 2 Ans: D

24. A precursor in the synthesis of the peptidoglycan of bacterial cell walls is UDP-:

A) galactose.
B) glucose.
C) glucuronic acid.
D) *N*-acetylglucosamine.
E) penicillin.

Synthesis of cell wall polysaccharides: plant cellulose and bacterial peptidoglycan
Page: 824 Difficulty: 2 Ans: D

25. Penicillin inhibits the synthesis of peptidoglycan:

A) branches.
B) chains.
C) crosslinks.
D) precursors.
E) All of the above

Synthesis of cell wall polysaccharides: plant cellulose and bacterial peptidoglycan
Page: 824 Difficulty: 2 Ans: C

26. Which one of the following reactions, cycles, or pathways is *not* found in plant systems?

A) The Calvin cycle
B) The gluconeogenesis pathway
C) The glyoxalate cycle
D) The rubisco reaction
E) The urea cycle

Integration of carbohydrate metabolism in the plant cell
Pages: 825–826 Difficulty: 1 Ans: E

27. Which one of the following cellular organelles is *not* unique to plant cells?

A) Amyloplasts
B) Chloroplasts
C) Glyoxysomes
D) Mitochondria
E) Vacuoles

Integration of carbohydrate metabolism in the plant cell
Pages: 825–826 Difficulty: 1 Ans: D

28. Which one of the following sugar phosphates is *not* part of the pool of readily interconvertible metabolites used by the plant cell?

    A) Dihydroxyacetone phosphate
    B) Fructose 2,6-bisphosphate
    C) Glucose 1-phosphate
    D) 6-phosphogluconate
    E) Xylulose 5-phosphate

Integration of carbohydrate metabolism in the plant cell
Pages: 825–826 Difficulty: 2 Ans: B

29. When glycerol is converted to glucose via gluconeogenesis in germinating seeds, the first glycolytic intermediate formed is:

    A) 1,3-bisphosphoglycerate.
    B) dihydroxyacetone phosphate.
    C) glycerol 1,3-bisphosphate.
    D) glycerol 3-phosphate.
    E) ribulose 1,5-bisphosphate.

Integration of carbohydrate metabolism in the plant cell
Pages: 825–826 Difficulty: 3 Ans: B

Short Answer Questions

30. Show the reaction catalyzed by ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco).

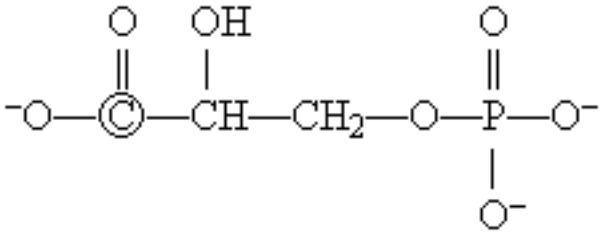
    Ans: Ribulose 1,5-bisphosphate + $CO_2 \longrightarrow$ 2 (3-phosphoglycerate)

    (For the details of the reaction, see Fig. 20-7.)

Photosynthetic carbohydrate synthesis
Page: 8 Difficulty: 3

31. Draw the structure of 3-phosphoglycerate. Circle the atom(s) that would be labeled first in plants grown in CO_2 labeled with radioactive carbon.

    Ans:



Photosynthetic carbohydrate synthesis

Pages: 801–803 Difficulty: 2

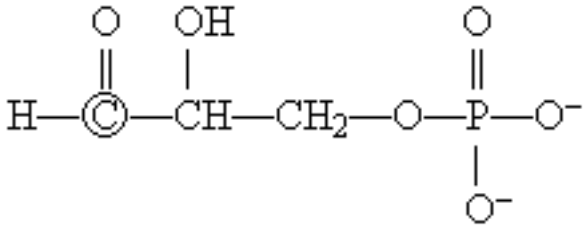
32. Show the reaction in which 3-phosphoglycerate is converted into glyceraldehyde 3-phosphate. Show all required cofactors, and circle the carbon atom(s) in glyceraldehyde 3-phosphate that is (are) derived from CO_2 during the photosynthetic fixation of CO_2.

Ans:

(1) 3-phosphoglycerate + ATP $\longrightarrow$ 1,3-bisphosphoglycerate + ADP

(2) 1,3-bisphosphoglycerate + NADPH + H^+ $\longrightarrow$ glyceraldehyde 3-phosphate + $NADP^+$ + P_i

The carbon atom at C-1 of glyceraldehyde 3-phosphate is derived from CO_2.



Photosynthetic carbohydrate synthesis
Page: 805 Difficulty: 3

33. Diagram the reaction catalyzed by transketolase when fructose 6-phosphate and glyceraldehyde 3-phosphate are the substrates.

Ans: Fructose 6-phosphate + glyceraldehyde 3-phosphate $\longrightarrow$
xylulose 5-phosphate + erythrose 4-phosphate

(See Fig. 20-11 for structures.)

Photosynthetic carbohydrate synthesis
Page: 807 Difficulty: 2

34. Explain why both ATP and NADPH are required for the operation of the Calvin cycle, and why these two reactants are required in different amounts.

Ans: ATP is required in two reactions of the Calvin cycle: the formation of 1,3-bisphosphoglycerate from 3-phosphoglycerate (2 ATP per CO_2 fixed) and the conversion of ribulose 5-phosphate into ribulose 1,5-bisphosphate (1 ATP per CO_2 fixed). NADPH is required in the reduction of 1,3-bisphosphoglycerate to glyceraldehyde 3-phosphate (2 NADPH per CO_2 fixed). Thus the operation of the cycle consumes 3 ATP and 2 NADPH per CO_2 fixed. (See Fig. 20-14, p. 782.)

Photosynthetic carbohydrate synthesis
Pages: 808–809 Difficulty: 2

35. How does glyceraldehyde 3-phosphate formed in the chloroplast stroma by the Calvin cycle reactions enter the cytosol?

Ans: It is converted to dihydroxyacetone phosphate or to 3-phosphoglycerate and carried into the cytosol by the P_i-triose phosphate antiporter, a specific transporter in the inner chloroplast membrane. (See Fig. 20-15.)

Photosynthetic carbohydrate synthesis
Pages: 809–810 Difficulty: 2

36. Describe how thioredoxin participates in the regulation of several chloroplast enzymes by light.

    Ans: Thioredoxin is an electron-carrying protein that is reduced by electrons from ferredoxin during illumination. Electrons from thioredoxin reduce critical disulfide bonds in key enzymes of the Calvin cycle, activating those enzymes. (See Fig. 20-19.)

Photosynthetic carbohydrate synthesis
Pages: 811–812 Difficulty: 2

37. Explain why photorespiration occurs in plant cells carrying out photosynthesis.

    Ans: Because rubisco is not particularly specific, it sometimes uses reacts with O_2, which is much more abundant than the proper substrate CO_2. This results in the production of 2-phosphoglycolate, which is salvaged by the glycolate pathway wherein O_2 is consumed and CO_2 is formed.

Photorespiration and the C_4 and CAM pathways
Pages: 812–813 Difficulty: 1

38. Describe the oxygenase activity of ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco) and explain why this reaction is undesirable from the point of view of a plant.

    Ans: The condensation of molecular oxygen with ribulose 1,5-bisphosphate yields 3-phosphoglycerate and the two-carbon compound 2-phosphoglycolate. 2-Phosphoglycolate has no known metabolic role; its carbon is salvaged by a series of reactions that consume O_2 and produce CO_2 – the "photorespiration" process. The salvage pathway requires energy, and therefore the oxygenase reaction of rubisco represents a net energy cost to the plant cell in which it occurs.

Photorespiration and the C_4 and CAM pathways
Pages: 812–813 Difficulty: 2

39. Describe the reaction sequence by which 2-phosphoglycolate (produced when O_2 replaces CO_2 as substrate for rubisco) is converted to serine. Name each enzyme and any cofactors required and indicate the subcellular compartment in which the reaction takes place.

    Ans: 2-phosphoglycolate is converted to glycolate by a phosphatase in the chloroplast. Glycolate is transported to the peroxisome and converted to glyoxylate by glycolate oxidase. The glyoxylate is then converted in the peroxisome to glycine by a transaminase that requires pyridoxal phosphate. Finally, two molecules of glycine are converted to serine + NH_3 + CO_2 by the enzyme glycine decarboxylase, which is located in the mitochondrion.

Photorespiration and the C_4 and CAM pathways
Page: 813 Difficulty: 3

40. CAM plants, such as cactus and pineapple, are native to very hot and dry environments. Briefly describe the biochemical events that allow CAM plants to minimize water loss by closing their stroma during daylight hours.

Ans: CAM plants fix CO_2 into malate in the dark when the stroma are open. The resulting malate is stored in vacuoles. During daylight hours, the CO_2 is released from malate by the action of the NADP-linked malic enzyme, and the CO_2 serves as substrate for rubisco.

Photorespiration and the C_4 and CAM pathways
Page: 815 Difficulty: 2

41. Diagram the pathway by which sucrose is synthesized from glucose 6-phosphate; indicate how any required cofactors are involved.

Ans: (See Fig. 20-24.) Glucose 6-phosphate is converted into glucose 1-phosphate by phosphoglucomutase, and into fructose 6-phosphate by phosphohexose isomerase. Then,

$$\text{Glucose 1-phosphate} + \text{UTP} \longrightarrow \text{UDP-glucose} + PP_i$$
$$\text{UDP-glucose} + \text{fructose 6-phosphate} \longrightarrow \text{sucrose 6-phosphate} + \text{UDP}$$
$$\text{Sucrose 6-phosphate} \longrightarrow \text{sucrose} + P_i$$

Biosynthesis of starch and sucrose
Page: 819 Difficulty: 3

42. Explain the utility to plants in using sucrose as the transport form of carbon.

Ans: Sucrose is a nonreducing disaccharide, with its glycosidic linkage between the anomeric carbons of glucose and fructose. The unavailability of these normally reactive carbons prevents sucrose from reacting nonenzymatically with amino acids or proteins, and this unusual bond is not hydrolyzed by amylases or other common carbohydrate-cleaving enzymes.

Biosynthesis of starch and sucrose
Pages: 819 Difficulty: 2

43. Describe briefly how the allosteric effector fructose 2,6-bisphosphate (F2,6BP) regulates starch and sucrose synthesis.

Ans: In vascular plants, the concentration of F2,6BP varies inversely with the rate of photosynthesis. At high levels of photosynthesis, the high levels of dihydroxyacetone phosphate that are formed inhibit phosphofructo-kinase-2. Thus, less F2,6BP is formed, which then leads to a greater flux of triose phosphate into fructose 6-phosphate formation and sucrose synthesis.

Biosynthesis of starch and sucrose
Page: 820 Difficulty: 2

44. Describe the role of fructose 2,6-bisphosphate in the regulation of sucrose biosynthesis in plant cells.

Ans: In conditions of intense light, 3-phosphoglycerate and dihydroxyacetone phosphate produced by the Calvin cycle inhibit PFK-2, slowing the conversion of fructose 6-phosphate to fructose 2,6-bisphosphate. The fructose 6-phosphate is then used to synthesize sucrose. In the dark, the fructose 2,6-bisphosphate level rises. Elevated levels of fructose 2,6-bisphosphate inhibit FBPase-1 and activate PFK-1, which results in an increase in the level of fructose 1,6-bisphosphate and hence increased glycolysis. (See Fig. 20-25.)

Biosynthesis of starch and sucrose
Page: 820 Difficulty: 3

45. Name one similarity and one difference between the biosynthesis of glycogen and the biosynthesis of cellulose.

    Ans: Similarities include the use of UDP-glucose as a precursor and a final composition containing only glucose. Differences are much more numerous and include a complex export and extracellular deposition mechanism for cellulose, lipid-linked intermediates, and the lack of branching in cellulose.

Synthesis of cell wall polysaccharides: plant cellulose and bacterial peptidoglycan
Pages: 821–822 Difficulty: 2

46. Describe how plants and some microorganisms can, unlike animals, convert acetyl-CoA derived from fatty acids into glucose or sucrose.

    Ans: The integration of reaction sequences in three subcellular compartments is required for the production of fructose 6-phosphate or sucrose from stored lipids. Some of the essential enzymes are sequestered in glyoxysomes, where glyoxysome-specific isozymes of β-oxidation break down fatty acids to acetyl-CoA. The physical sepration of the glyoxylate cycle and β-oxidation enzymes from the mitochondrial citric acid cycle enzymes prevents further oxidation of acetyl-CoA to CO_2. Instead, the acetyl-CoA is converted to succinate in the glyoxylate cycle, which passes into the mitochondrial matrix, where it is converted by citric acid cycle enzymes to oxaloacetate, which moves into the cytosol. Cyotsolic oxaloacetate is converted by gluconeogenesis to fructose 6-phosphate, a precursor for glucose or sucrose.

Integration of carbohydrate metabolism in the plant cell
Pages: 825–826 Difficulty: 3

47. Describe the interconversions between the triose-, pentose- and hexose-phosphate pools in the plant cell.

    Ans: Compounds in each pool are readily interconverted by enzymatic reactions with small standard free-energy changes. Thus, a temporary imbalance is rapidly corrected to achieve a new equilibrium state. Specific interconversions are depicted in Fig. 20-34.

Integration of carbohydrate metabolism in the plant cell
Page: 827 Difficulty: 2